

Tristan Louis

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EDUCATION

Queens College
Bachelor of Science in Computer Science

65-30 Kissena Blvd., Flushing, NY 11367
Graduated: May 2025

CLUBS & LEADERSHIP

- Weight Room Club—Active member focused on strength training, fitness, and discipline.
- Queens College Stock Broker Exchange Club—Engaged in financial markets, trading strategies, and investment analysis.
- Queens College CODE FOR ALL—Contributed to coding initiatives promoting accessibility and inclusivity in tech.
- D.I.C.E. (Diversity in Computing and Engineering)—Advocated for diversity and inclusion in STEM fields.

EXPERIENCE

Data Science Intern

Dakdan Worldwide- Remote

August 2025 - Present

- Process and analyze large sports-business datasets using **Python, SQL, and Excel** to extract insights for sales and marketing strategy.
- Build automated **ETL pipelines and visualization dashboards, integrating external data (Google Trends, Twitter, Twitch, etc)** with internal metrics.
- **Deliver client-facing reports and statistical analyses**, translating findings into actionable strategies that support revenue growth.

Software Development Intern

OpenQQuantify - Remote

June 2025 - Present

- Built full-stack features using **Flask, HTML/CSS, JavaScript, and SQLAlchemy**, including **REST APIs** and **LLM-powered endpoints**.
- Actively improving technical proficiency in **low-level systems, backend logic, and frontend tools** in a hands-on, community-driven engineering environment.
- Helped prototype **AI-integrated 3D simulations** with CesiumJS and contributed to early **digital twin and electronics pipeline integration**.

Data Engineer

Open Avenues Foundation: The Build Fellowship Highlight Reel: Automated Sports Content Generation

September 2024 – December 2024

- Spearheaded the development of an AI-driven sports highlight generator, **leveraging machine learning & computer vision**.
- Engineered a real-time video processing pipeline using **XGBoost** and **OpenCV**, integrating **ball tracking and automated transitions**.
- Refined models through iterative testing, **improving highlight accuracy by 12%** and reducing false positives in play detection.
- Collaborated with engineers and industry mentors, implementing feature engineering techniques to **enhance model performance**.

PROJECTS

Predictive Analytics for Diabetes Management

Pandas, AWS SageMaker, AWS Wrangler, NumPy, scikit-learn (KMeans, PCA, StandardScaler), mlxtend (Apriori, association rules), Matplotlib

- Conducted **machine learning-based analysis** of diabetes treatment patterns using **classification models and clustering techniques**.
- Applied the **Apriori algorithm** for **association rule mining**, identifying key relationships between **A1C levels, medication changes, and emergency visits**.
- Developed **classification models** to predict patient outcomes, achieving **85.3% accuracy in medication change predictions** and **90.2% accuracy in emergency visit forecasting**.
- Performed **clustering analysis** to segment patients based on treatment plans, providing insights into personalized diabetes care.

Customer Credit Risk Prediction Using Machine Learning

Kaggle, Pandas, NumPy, scikit-learn (Logistic Regression, Decision Tree, Random Forest), XGBoost, Matplotlib, Seaborn, GridSearchCV, SMOTE

- Built a **credit risk assessment model** to predict **loan default probabilities** using **XGBoost & Random Forest**.
- Engineered financial features like **debt-to-income ratio, credit utilization, and payment history trends**, improving accuracy to **88%**.
- Implemented **SHAP (Shapley Additive Explanations)** for explainability, providing financial analysts with insights into **risk factors**.
- Built an **interactive dashboard using Streamlit**, enabling lenders to assess borrower risk in real time and make data-driven loan decisions.

PathMark Interactive BOT

Python, JSON, Flask, Gemini API, SpeechRecognition, pyttsx3, PyAudio

- Developed a **voice-activated assistant** that helps users find grocery items by querying a **nested JSON** inventory using natural language.
- Enabled customers to locate items via **keyword or voice input** and receive **audio responses**
- Planning the integration of **NLP and voice input** for future versions

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, Java, C++, HTML/CSS, JavaScript
- **Machine Learning & AI:** XGBoost, OpenCV, TensorFlow, Feature Engineering, Data Preprocessing
- **Data Processing & Visualization:** Pandas, NumPy, Scikit-learn, Matplotlib
- **Databases & Backend:** MySQL, PostgreSQL, MongoDB, REST APIs
- **Cloud & Deployment:** Google Cloud Platform, Flask, FastAPI, Streamlit
- **Developer Tools:** Git, Linux, VS Code, PyCharm, IntelliJ

CERTIFICATIONS

- **Build Fellowship Completion**—An AI-powered sports highlight generation using ML & computer vision.
- **Code Path Intermediate Technical Prep (TIP102)**—Python, data structures, and algorithms
- **Udemy Full-Stack Web Development Bootcamp**—Node.js, Express, React, and database management.
- **Udemy Python for Data Science & Machine Learning**—Python, data structures, algorithms, and AI applications